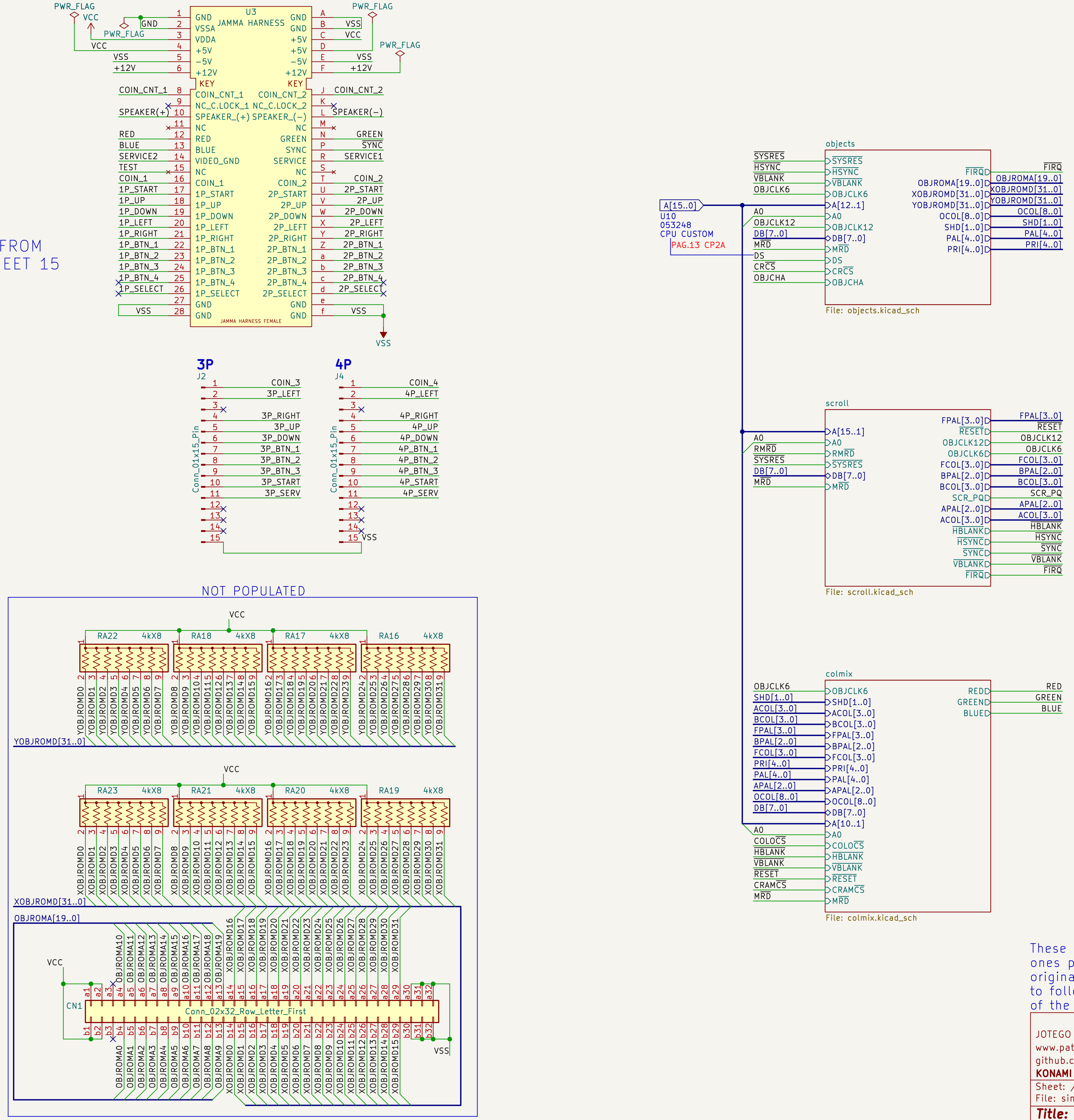
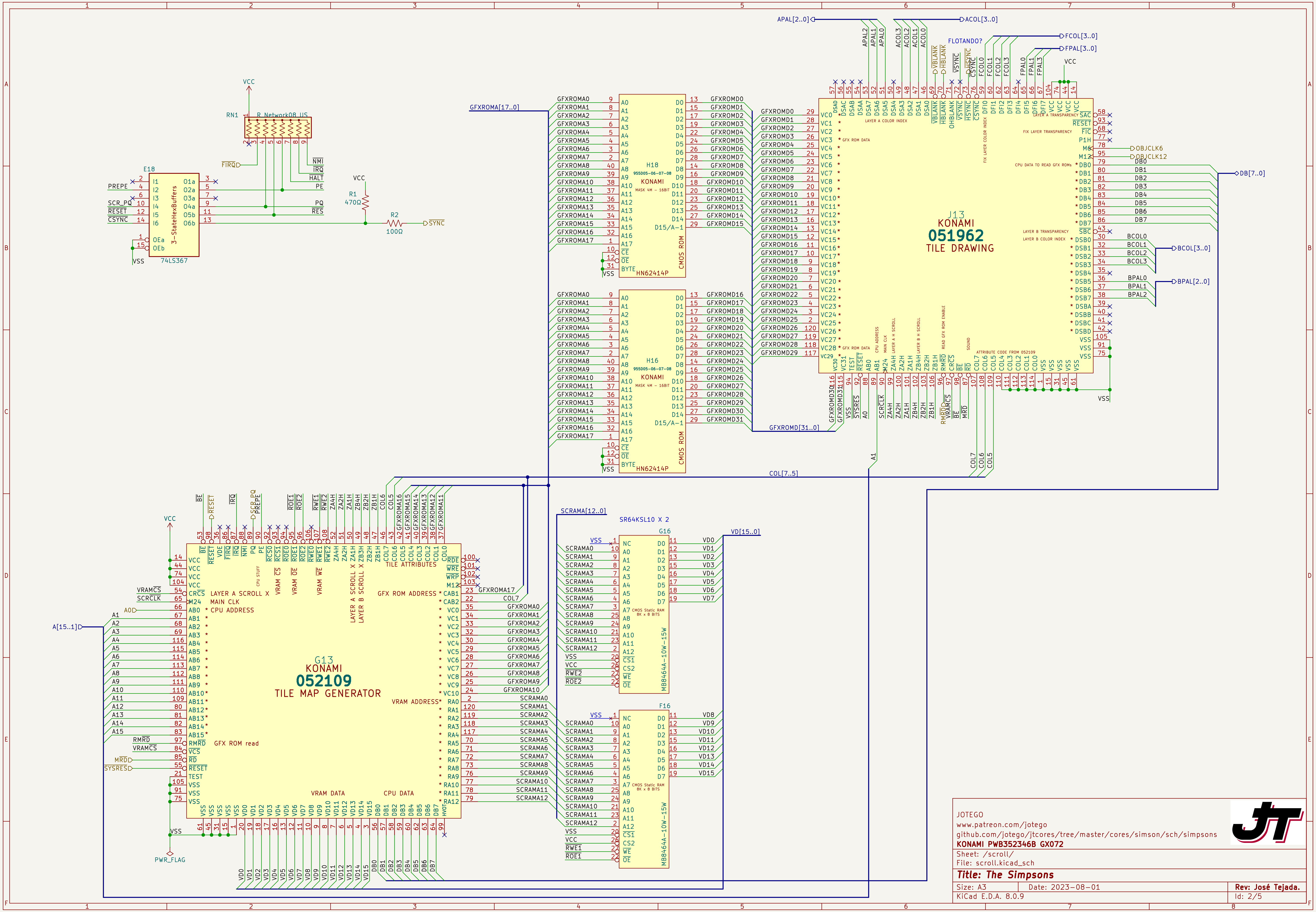
THESE SIGNALS COME FROM
SCH SIMPSONS I/O SHEET 15


U3
JAMMA HARNESS
JAMMA HARNESS FEMALE
PWR_FLAG
VCC
GND
VSSA
VDDA
+5V
-5V
+12V
VSS
KEY
COIN_CNT_1
COIN_CNT_2
NC_C.LOCK_1
NC_C.LOCK_2
SPEAKER_(+)
SPEAKER_(-)
SPEAKER(+)
SPEAKER(-)
NC
RED
GREEN
BLUE
SYNC
VIDEO_GND
SERVICE
SERVICE1
SERVICE2
TEST
COIN_1
COIN_2
1P_START
2P_START
1P_UP
2P_UP
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
1P_BTN_4
2P_BTN_4
1P_SELECT
2P_SELECT
3P
J2
Conn_01x15_Pin
COIN_3
3P_LEFT
3P_RIGHT
3P_UP
3P_DOWN
3P_BTN_1
3P_BTN_2
3P_BTN_3
3P_START
3P_SERV
4P
J4
COIN_4
4P_LEFT
4P_RIGHT
4P_UP
4P_DOWN
4P_BTN_1
4P_BTN_2
4P_BTN_3
4P_START
4P_SERV
NOT POPULATED
RA22 4kX8
RA18 4kX8
RA17 4kX8
RA16 4kX8
RA23 4kX8
RA21 4kX8
RA20 4kX8
RA19 4kX8
YOBJROMD0 … YOBJROMD31
YOBJROMD[31..0]
XOBJROMD0 … XOBJROMD31
XOBJROMD[31..0]
OBJROMA[19..0]
OBJROMA0 … OBJROMA19
CN1
Conn_02x32_Row_Letter_First
A[15..0]
U10
053248
CPU CUSTOM
PAG.13 CP2A
objects
SYSRES
HSYNC
VBLANK
OBJCLK6
A[12..1]
A0
OBJCLK12
DB[7..0]
MRD
DS
CRCS
OBJCHA
FIRQ
OBJROMA[19..0]
XOBJROMD[31..0]
YOBJROMD[31..0]
OCOL[8..0]
SHD[1..0]
PAL[4..0]
PRI[4..0]
File: objects.kicad_sch
scroll
A[15..1]
RMRD
FPAL[3..0]
RESET
OBJCLK12
OBJCLK6
FCOL[3..0]
BPAL[2..0]
BCOL[3..0]
SCR_PQ
APAL[2..0]
ACOL[3..0]
HBLANK
HSYNC
SYNC
VBLANK
FIRQ
File: scroll.kicad_sch
colmix
OBJCLK6
SHD[1..0]
ACOL[3..0]
BCOL[3..0]
FPAL[3..0]
BPAL[2..0]
FCOL[3..0]
PRI[4..0]
PAL[4..0]
APAL[2..0]
OCOL[8..0]
DB[7..0]
A[10..1]
A0
COLOCS
HBLANK
VBLANK
RESET
CRAMCS
MRD
RED
GREEN
BLUE
File: colmix.kicad_sch

These schematics cover the missing pages in the original ones published by the maker. Signal names existing in the original pages have been preserved. New signal names try to follow the maker style or are taken from similar schematics of the same maker.

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
**KONAMI PWB352346B GX072**
Sheet: /
File: simpsons.kicad_sch
***Title: The Simpsons***
Size: A3 Date: 2023-08-01 **Rev: José Tejada.**
KiCad E.D.A. 8.0.9 Id: 1/5


E18
74LS367
3-StateHexBuffers
RN1
R_Network08_US
R1
470Ω
R2
100Ω
VCC
VSS
PREPE
SCR_PQ
RESET
CSYNC
FIRQ
NMI
IRQ
HALT
PE
PQ
RES
SYNC
H18
H16
955005-06-07-08
KONAMI
MASK 4M - 16BIT
CMOS ROM
HN62414P
GFXROMA[17..0]
GFXROMD[31..0]
J13
KONAMI
051962
TILE DRAWING
APAL[2..0]
ACOL[3..0]
FCOL[3..0]
FPAL[3..0]
FLOTANDO?
OBJCLK6
OBJCLK12
DB[7..0]
BCOL[3..0]
BPAL[2..0]
LAYER A COLOR INDEX
FIX LAYER COLOR INDEX
FIX LAYER TRANSPARENCY
LAYER A TRANSPARENCY
CPU DATA TO READ GFX ROMs
LAYER B TRANSPARENCY
LAYER B COLOR INDEX
GFX ROM DATA
CPU ADDRESS
MAIN CLK
LAYER A H SCROLL
LAYER B H SCROLL
READ GFX ROM ENABLE
SOUND
ATTRIBUTE CODE FROM 052109
COL[7..5]
SCRAMA[12..0]
SR64KSL10 X 2
G16
F16
CMOS Static RAM 8K x 8 BITS
MB8464A-10W-15W
VD[15..0]
G13
KONAMI
052109
TILE MAP GENERATOR
LAYER A SCROLL X
MAIN CLK
CPU ADDRESS
CPU STUFF
VRAM CS
VRAM OE
VRAM WE
LAYER A SCROLL X
LAYER B SCROLL X
TILE ATTRIBUTES
GFX ROM ADDRESS
VRAM ADDRESS
GFX ROM read
VRAM DATA
CPU DATA
A[15..1]
A0
MRD
SYSRES
PWR_FLAG
VRAMCS
SCRCLK
RMRD

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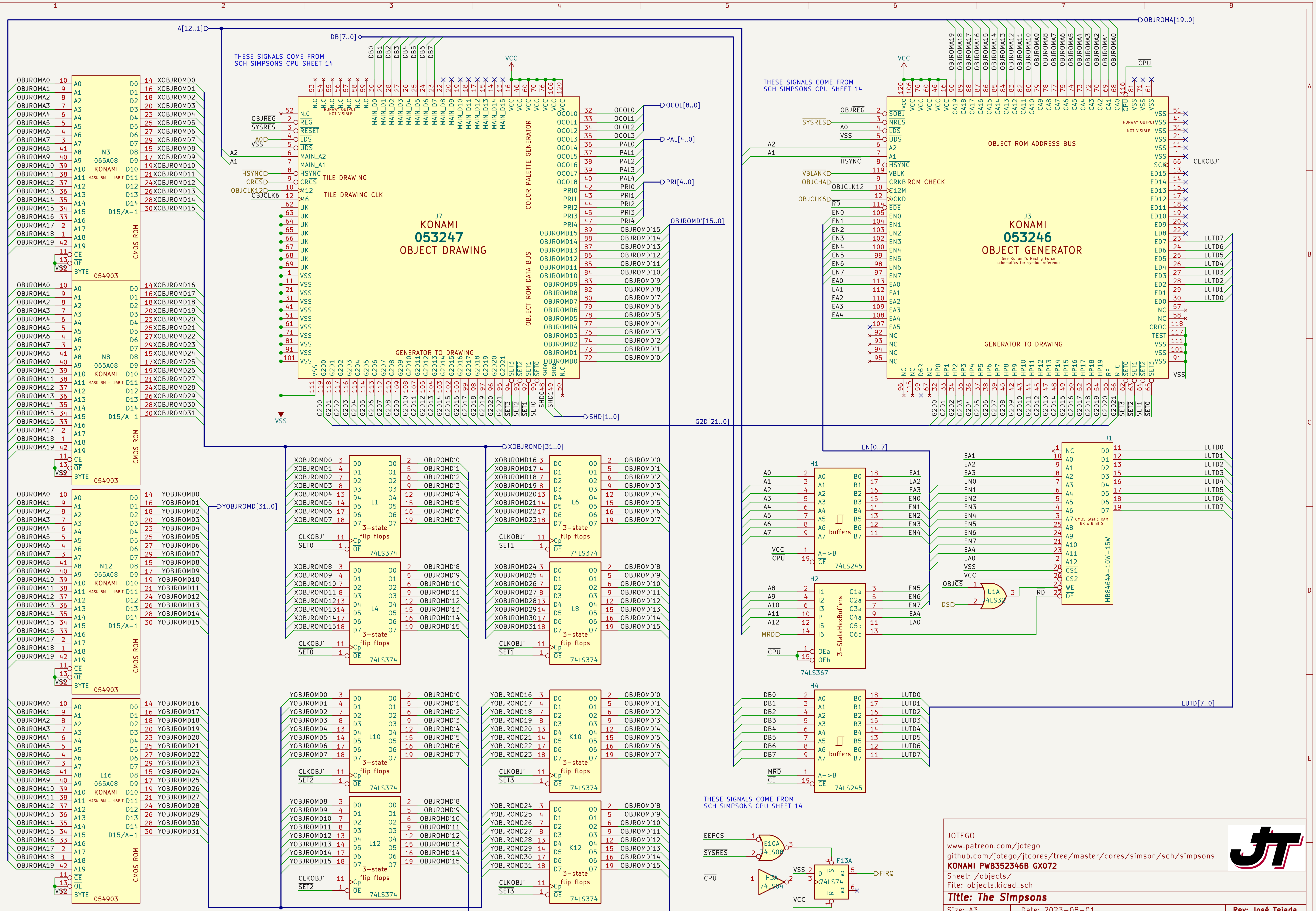
Sheet: /scroll/
File: scroll.kicad_sch

**Title: The Simpsons**

| Size: A3 | Date: 2023-08-01 | **Rev: José Tejada.** |
|---|---|---|
| KiCad E.D.A. 8.0.9 | | Id: 2/5 |


THESE SIGNALS COME FROM SCH SIMPSONS CPU SHEET 14
A[12..1]
DB[7..0]
OBJROMA[19..0]
VCC
VSS
N3 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
N8 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
N12 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
L16 065A08 KONAMI MASK 8M - 16BIT CMOS ROM 054903
J7 KONAMI 053247 OBJECT DRAWING
RUNWAY OUTPUT NOT VISIBLE
TILE DRAWING
TILE DRAWING CLK
COLOR PALETTE GENERATOR
OBJECT ROM DATA BUS
GENERATOR TO DRAWING
OCOL[8..0]
PAL[4..0]
PRI[4..0]
OBJROMD'[15..0]
SHD[1..0]
G2D[21..0]
XOBJROMD[31..0]
YOBJROMD[31..0]
J3 KONAMI 053246 OBJECT GENERATOR
See Konami's Racing Force schematics for symbol reference
OBJECT ROM ADDRESS BUS
ROM CHECK
CLKOBJ'
EN[0..7]
LUTD[7..0]
L1 74LS374 3-state flip flops
L4 74LS374 3-state flip flops
L6 74LS374 3-state flip flops
L8 74LS374 3-state flip flops
L10 74LS374 3-state flip flops
L12 74LS374 3-state flip flops
K10 74LS374 3-state flip flops
K12 74LS374 3-state flip flops
H1 74LS245 buffers
H2 74LS367 3-StateHexBuffers
H4 74LS245 buffers
J1 MB8464A-10W-15W CMOS Static RAM 8K x 8 BITS
U1A 74LS32
E10A 74LS08
H3A 74LS04
F13A 74LS74
EEPCS
SYSRES
CPU
FIRQ
OBJCS
DSD
RD

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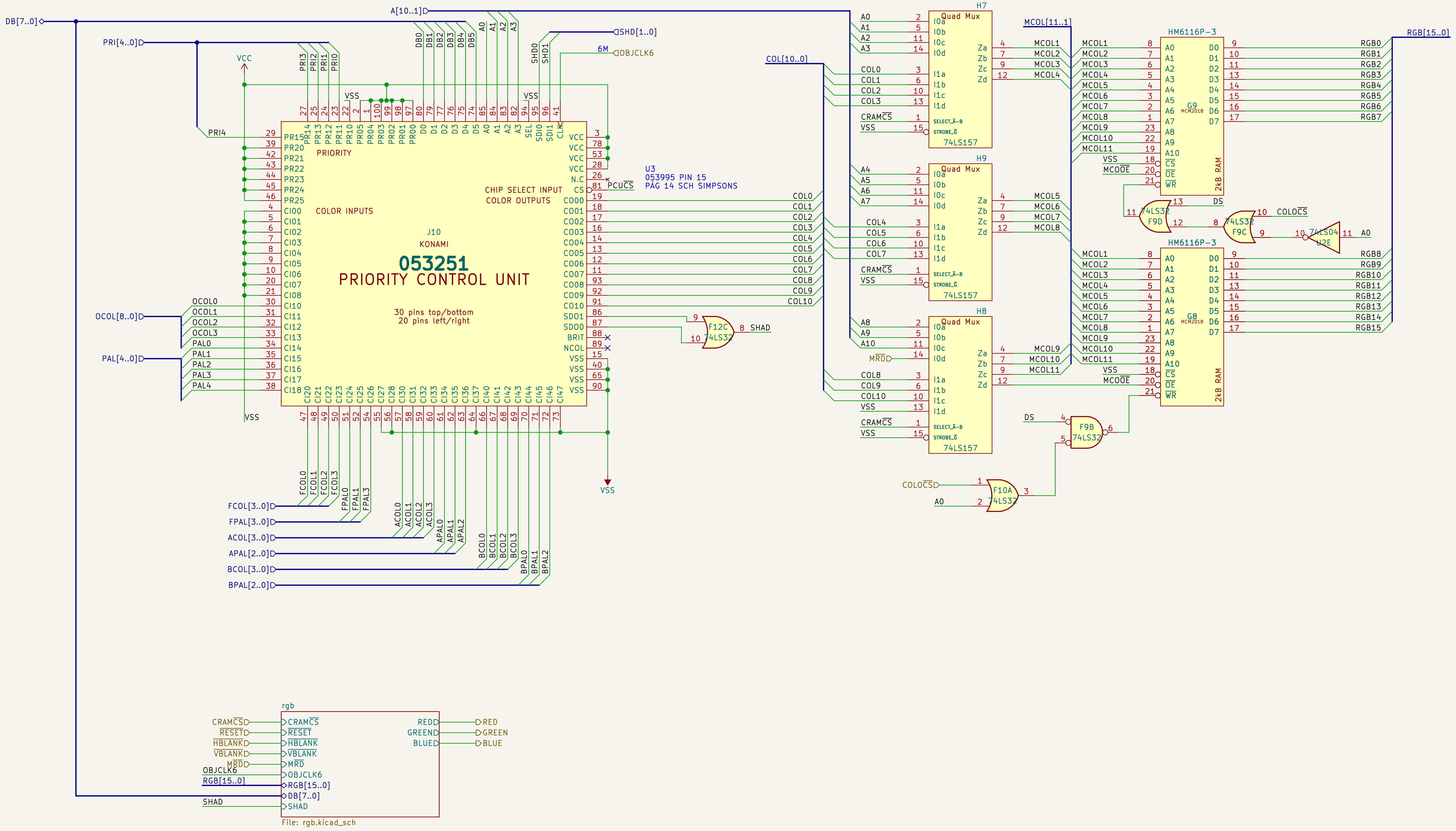
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
**KONAMI PWB352346B GX072**

Sheet: /objects/
File: objects.kicad_sch

**Title: The Simpsons**

Size: A3 Date: 2023-08-01 **Rev: José Tejada.**
KiCad E.D.A. 8.0.9 Id: 3/5


DB[7..0]
PRI[4..0]
A[10..1]
SHD[1..0]
OBJCLK6
6M
VCC
VSS
PRI4
PRI3
PRI2
PRI1
PRI0
DB0
DB1
DB2
DB3
DB4
DB5
A0
A1
A2
A3
SHD0
SHD1
J10
KONAMI
053251
PRIORITY CONTROL UNIT
30 pins top/bottom
20 pins left/right
PRIORITY
COLOR INPUTS
CHIP SELECT INPUT
COLOR OUTPUTS
U3
053995 PIN 15
PÁG 14 SCH SIMPSONS
PCUCS
OCOL[8..0]
PAL[4..0]
OCOL0
OCOL1
OCOL2
OCOL3
PAL0
PAL1
PAL2
PAL3
PAL4
COL0
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
COL9
COL10
F12C
74LS32
SHAD
FCOL[3..0]
FPAL[3..0]
ACOL[3..0]
APAL[2..0]
BCOL[3..0]
BPAL[2..0]
FCOL0
FCOL1
FCOL2
FCOL3
FPAL0
FPAL1
FPAL3
ACOL0
ACOL1
ACOL2
ACOL3
APAL0
APAL1
APAL2
BCOL0
BCOL1
BCOL2
BCOL3
BPAL0
BPAL1
BPAL2
COL[10..0]
MCOL[11..1]
H7
H9
H8
Quad Mux
74LS157
CRAMCS
MRD
HM6116P-3
G9
G8
MCM2018
2kB RAM
MCOE
RGB[15..0]
RGB0
RGB1
RGB2
RGB3
RGB4
RGB5
RGB6
RGB7
RGB8
RGB9
RGB10
RGB11
RGB12
RGB13
RGB14
RGB15
DS
F9D
F9C
U2E
74LS04
COLOCS
F9B
F10A
A0
rgb
CRAMCS
RESET
HBLANK
VBLANK
MRD
OBJCLK6
RGB[15..0]
DB[7..0]
SHAD
REDD
GREEND
BLUED
RED
GREEN
BLUE
File: rgb.kicad_sch

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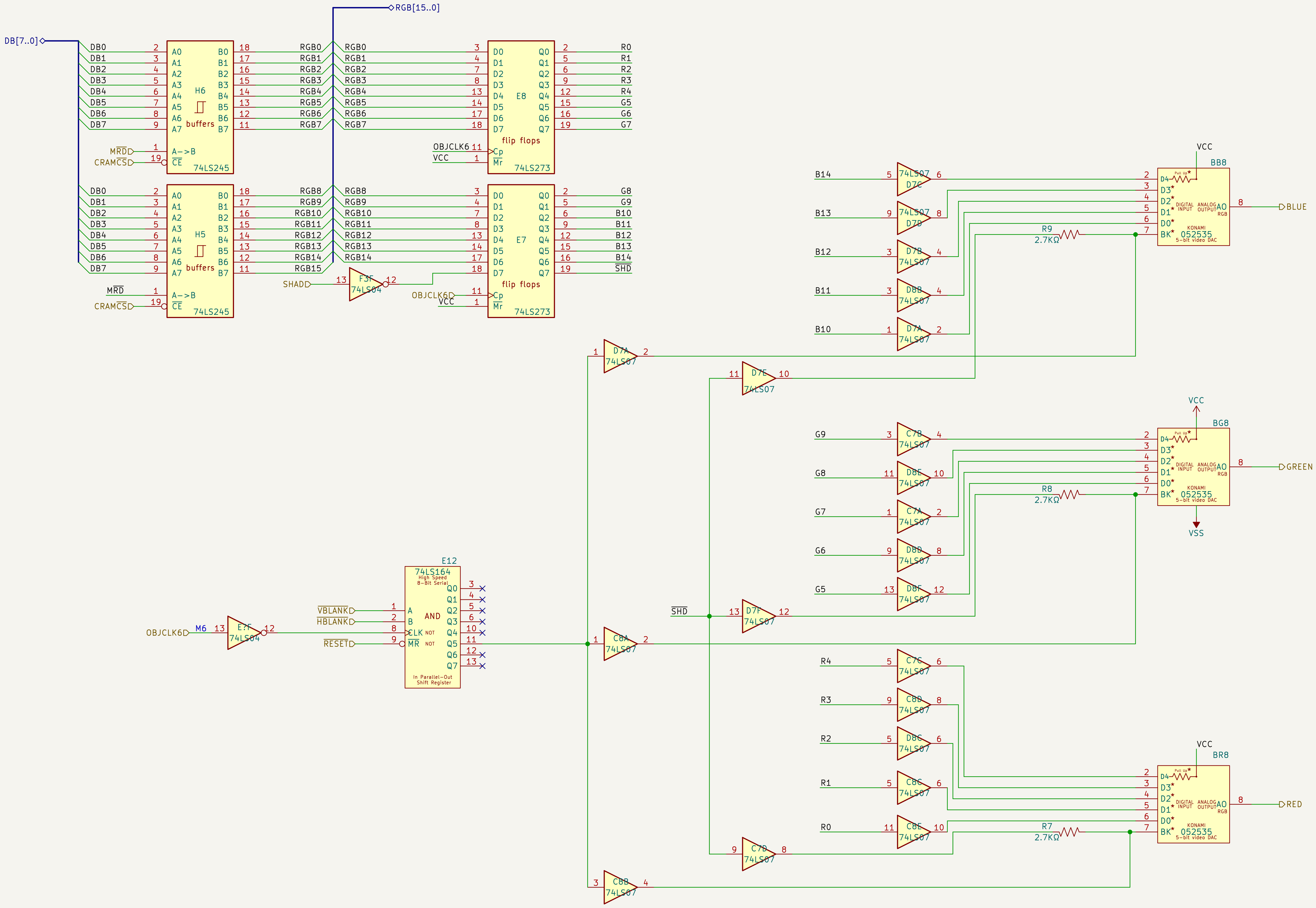
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
**KONAMI PWB352346B GX072**
Sheet: /colmix/
File: colmix.kicad_sch

**Title: The Simpsons**

Size: A3 Date: 2023-08-01 **Rev: José Tejada.**
KiCad E.D.A. 8.0.9 Id: 4/5


RGB[15..0]
DB[7..0]
DB0 DB1 DB2 DB3 DB4 DB5 DB6 DB7
H6
buffers
74LS245
A0 A1 A2 A3 A4 A5 A6 A7
B0 B1 B2 B3 B4 B5 B6 B7
A->B
CE
MRD
CRAMCSD
RGB0 RGB1 RGB2 RGB3 RGB4 RGB5 RGB6 RGB7
E8
flip flops
74LS273
D0 D1 D2 D3 D4 D5 D6 D7
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
Cp
Mr
OBJCLK6
VCC
R0 R1 R2 R3 R4 G5 G6 G7
H5
RGB8 RGB9 RGB10 RGB11 RGB12 RGB13 RGB14 RGB15
MRD
CRAMCSD
SHADD
F3F
74LS04
E7
OBJCLK6D
G8 G9 B10 B11 B12 B13 B14 SHD
B14
B13
B12
B11
B10
D7C
D7D
D7B
D8B
D7A
74LS07
VCC
BB8
Pull Up
D4 D3 D2 D1 D0 BK
DIGITAL INPUT
ANALOG OUTPUT
A0
RGB
KONAMI
052535
5-bit video DAC
BLUE
R9
2.7KΩ
D7E
G9
G8
G7
G6
G5
C7D
D8E
C7A
D8D
D8F
BG8
GREEN
VSS
R8
2.7KΩ
SHD
D7F
R4
R3
R2
R1
R0
C7C
C8D
D8C
C8C
C8E
BR8
RED
R7
2.7KΩ
C7B
C8A
C8B
E12
74LS164
High Speed 8-Bit Serial
In Parallel-Out Shift Register
A
B
AND
CLK
MR
NOT
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
VBLANKD
HBLANKD
RESETD
OBJCLK6D
M6
E7F
74LS04

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**KONAMI PWB352346B GX072**
Sheet: /colmix/rgb/
File: rgb.kicad_sch

**Title: The Simpsons**

| | | |
|---|---|---|
| Size: A3 | Date: 2023-08-01 | **Rev: José Tejada.** |
| KiCad E.D.A. 8.0.9 | | Id: 5/5 |